

Digital Engineering - Assignment 3

Name: Abdullah Mohammed Ali Hassan

ID: 2420766

Group: G2C4

Question 1: Odd Number Function $F(A,B,C)$

1. Truth Table

Construct the truth table for the function $F(A, B, C)$ where $F = 1$ if the binary number represented by A, B, C is an odd number, and $F = 0$ otherwise.

Decimal	A	B	C	F
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1

Decimal	A	B	C	F
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	1

2. Minterms and Maxterms

From your truth table, determine all the minterms (where $F = 1$) and maxterms (where $F = 0$).

- Minterms ($F = 1$):

$$\Sigma(1, 3, 5, 7)$$

- Maxterms ($F = 0$):

$$\Pi(0, 2, 4, 6)$$

3. Karnaugh Map Simplification (SOP)

Simplify the function using a 3-variable Karnaugh Map (K-map) to obtain the minimal Sum of Products (SOP) form, draw the map, and show all groupings.

		BC			
		00	01	11	10
A	0	0 ^{m0}	1 ^{m1}	1 ^{m3}	0 ^{m2}
	1	0 ^{m4}	1 ^{m5}	1 ^{m7}	0 ^{m6}

4. Simplified Boolean Expression

Write the simplified Boolean expression for $F(A, B, C)$ from K-maps.

$$F(A, B, C) = C$$

5. Dual of the Simplified Function

Find the dual of the simplified function.

$$F^D = C$$

6. Complement of the Simplified Function

Find the complement of the simplified function.

$$\bar{F} = \bar{C}$$

7. Logic Circuit Diagram

Draw the logic circuit diagram for the simplified function using basic logic gates (AND, OR, NOT).

C ————— F

Question 2: Two or More 1-Bits Function F(A,B,C)

1. Truth Table

Construct the truth table for the function $F(A, B, C)$ where $F = 1$ if the number of 1-bits represented by A, B, C is 2 or more, and $F = 0$ otherwise.

Decimal	A	B	C	Count of 1s	F
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	2	1
4	1	0	0	1	0
5	1	0	1	2	1
6	1	1	0	2	1
7	1	1	1	3	1

2. Minterms and Maxterms

From your truth table, determine all the minterms (where $F = 1$) and maxterms (where $F = 0$).

- Minterms ($F = 1$):

$$\Sigma(3, 5, 6, 7)$$

- Maxterms ($F = 0$):

$$\Pi(0, 1, 2, 4)$$

3. Karnaugh Map Simplification (SOP)

Simplify the function using a 3-variable Karnaugh Map (K-map) to obtain the minimal Sum of Products (SOP) form, draw the map, and show all groupings.

		BC			
		00	01	11	10
A	0	0 ^{m0}	0 ^{m1}	1 ^{m3}	0 ^{m2}
	1	0 ^{m4}	1 ^{m5}	1 ^{m7}	1 ^{m6}

4. Simplified Boolean Expression

Write the simplified Boolean expression for $F(A, B, C)$ from K-maps.

$$F(A, B, C) = BC + AB + AC$$

5. Dual of the Simplified Function

Find the dual of the simplified function.

$$F^D = (B + C)(A + B)(A + C)$$

6. Complement of the Simplified Function

Find the complement of the simplified function.

$$\bar{F} = \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{C}$$

7. Logic Circuit Diagram

Draw the logic circuit diagram for the simplified function using basic logic gates (AND, OR, NOT).

